

Global Superstore Sales & Profitability Analysis

Tracing an 11.63% profit margin back to its root cause using SQL Server data cleaning and an interactive Power BI dashboard.

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Tools: SQL Server (T-SQL), Power BI Desktop

Dataset: Global Superstore (2011-2014), 51,291 order lines

\$12.64M

TOTAL SALES

\$1.47M

TOTAL PROFIT

14.26%

AVG DISCOUNT

11.63%

PROFIT MARGIN

12.18%

RETURN RATE

1. Executive Summary

Global Superstore generated \$12.64M in total sales across 2011-2014 but converted only 11.63% of it into profit. This report documents an end-to-end analysis — from raw, unaudited transactional data through SQL Server cleaning and modeling, to a single interactive Power BI dashboard — built to answer one question: where exactly is profit margin being lost, and why?

Bottom line: The erosion is concentrated, not spread evenly across the business. One sub-category (Tables), reinforced by an aggressive average discount of ~29%, is operating at a **-8.41% margin** and losing roughly **\$63.6K** outright — while regional performance and a 12.18% return rate compound the impact further.

2. Business Problem

Global Superstore's overall profit margin sits well below what a healthy retail portfolio should carry, despite strong top-line sales. Leadership needs to know which regions, product categories, and discounting behaviors are eroding profitability, and how much of that erosion is being made worse by product returns — so corrective action can be targeted rather than applied blindly across the whole business.

3. Data & Methodology

Table	Rows	Description
dbo.Orders	51,291	Fact table — order/ship dates, customer, location, product, sales metrics
dbo.People	14	Dimension — maps each Region to its Regional Manager
dbo>Returns	~800+	Flag table — Order IDs that were returned

EDA: row counts verified across all three tables; schema confirmed via `INFORMATION_SCHEMA.COLUMNS`; null audit uncovered 41 rows with missing Profit; duplicate checks confirmed Row_ID uniqueness and caught one duplicate Order_ID in Returns.

Cleaning: three governed views (vw_CleanOrders, vw_CleanReturns, vw_CleanPeople) sit on top of the raw tables — nothing is edited in place, so every transformation stays auditable. These are joined into vw_Final_Superstore_Dashboard, the single object Power BI connects to.

KPI Logic: four SQL queries answer the core business questions before any chart is built — headline company metrics, return impact, weakest-margin region, and weakest-margin sub-category.

4. Power BI Dashboard

The final SQL view was imported into Power BI (Import mode) and used to build DAX measures for Total Sales, Total Profit, Profit Margin %, Average Discount, and Return Rate %, alongside a single-page dashboard with an Is_Returned slicer to filter the entire page by return status.



Figure 1. Global Superstore Executive Dashboard — Power BI Desktop.

5. Key Findings

- Tables is the single biggest profit leak.** It is the only sub-category operating at a negative margin (-8.41%), losing approximately \$63.6K on \$755.9K in sales — while every other sub-category remains profitable, led by Paper at 24.29%.
- Discounting and margin move in opposite directions.** The scatter plot shows a clear negative relationship, and Tables sits at the extreme end with an average discount near 29% — the highest of any sub-category.
- Regional performance is uneven.** Canada leads at a 26.62% margin, while Africa (11.35%) and East (13.48%) trail behind the company average, with Southeast Asia the weakest performer company-wide at roughly 2.05%.
- Consumer is the most valuable segment,** contributing just over half of total profit (51.06%), ahead of Corporate (30.08%) and Home Office (18.86%) combined.
- Returns are a material cost.** A 12.18% return rate represents a meaningful share of sales and profit exposure — enough to justify its own filter on the dashboard.

6. Recommendations

- Immediate**

 - Cap or restructure discounting on the Tables sub-category — the clearest, most isolated driver of margin loss.
 - Review pricing and vendor cost for the specific loss-making Table products identified in the dashboard's product-level detail table.
- Strategic**

 - Investigate root causes of underperformance in Southeast Asia and Africa — cost structure, local pricing, or logistics.
 - Analyze return reasons for high-return categories to reduce the 12.18% return rate over time.

7. Conclusion

This project demonstrates a complete analyst workflow — from raw, unaudited data to a governed SQL model to an executive-ready Power BI dashboard — used to isolate a specific, actionable driver of margin erosion rather than leaving leadership with a single vague company-wide percentage. Correcting the discount practice on the Tables sub-category represents the highest-leverage, lowest-risk opportunity identified in this analysis to recover profit margin without reducing overall sales volume.